Michigan Tech

Two-Body System

Reduction of Newton's *n*-body equations

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Outline

1. Specialize Newton's n-body equations to two masses.
2. Separate center-of-mass motion from relative motion.
3. Derive the reduced mass and reconstruct both positions.
4. Apply the formulas to the Earth–Moon system.

Newton's n-body system

- Mass $m_i > 0$ at position $\boldsymbol{r}_i(t) \in \mathbb{R}^3$.
- Non-colliding isolated point masses obey (in inertial frames) Newton's laws [1].

$$m_i \boldsymbol{r}_i'' = -G \sum_{\substack{j=1 \\ j \neq i}}^{n} m_i m_j \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^3} \qquad i = 1, \ldots, n$$

- G is the Gravitational constant. Primes denote time derivatives.

Acceleration

- Divide equation i by m_i.

$$\boldsymbol{r}_i'' = -G \sum_{\substack{j=1 \\ j \neq i}}^{n} m_j \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^3}$$

  Acceleration depends on all other positions.
- $n = 2$ gives an isolated two-body system. Selecting two bodies from a larger system neglects forces from the remaining bodies.

Two-body equations

- For $n = 2$

$$\boldsymbol{r}_1'' = -Gm_2 \frac{\boldsymbol{r}_1 - \boldsymbol{r}_2}{\|\boldsymbol{r}_1 - \boldsymbol{r}_2\|^3}$$
$$\boldsymbol{r}_2'' = -Gm_1 \frac{\boldsymbol{r}_2 - \boldsymbol{r}_1}{\|\boldsymbol{r}_2 - \boldsymbol{r}_1\|^3}$$

- Forces are equal and opposite, scaling and adding gives

$$m_1 \boldsymbol{r}_1'' + m_2 \boldsymbol{r}_2'' = \boldsymbol{0}$$

Center-of-mass and relative coordinates

- Define total mass $M = m_1 + m_2$ and **Center of Mass**

$$\boldsymbol{c} = \frac{m_1 \boldsymbol{r}_1 + m_2 \boldsymbol{r}_2}{M}$$

- Define the **Relative Position** from body 1 to body 2

$$\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1 \qquad r = \|\boldsymbol{r}\| > 0$$

- $\boldsymbol{c}$ is the position of the pair. $\boldsymbol{r}$ is the displacement between the bodies.

Center-of-mass motion

1. Differentiate the mass-weighted position twice.

$$M\boldsymbol{c}'' = m_1\boldsymbol{r}_1'' + m_2\boldsymbol{r}_2'' = \boldsymbol{0}$$

2. Integrate the resulting equation.

$$\boldsymbol{c}(t) = \boldsymbol{c}(0) + t\boldsymbol{c}'(0)$$

3. The center of mass moves at constant velocity. Choose its rest frame to set $\boldsymbol{c} = \boldsymbol{0}$.

Relative motion

1. Use $\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1$ to rewrite the accelerations.

$$\boldsymbol{r}_1'' = Gm_2\frac{\boldsymbol{r}}{r^3} \qquad \boldsymbol{r}_2'' = -Gm_1\frac{\boldsymbol{r}}{r^3}$$

2. Subtract the first acceleration from the second.

$$\boldsymbol{r}'' = \boldsymbol{r}_2'' - \boldsymbol{r}_1'' = -G(m_1 + m_2)\frac{\boldsymbol{r}}{r^3}$$

3. Relative acceleration points toward $\boldsymbol{r} = \boldsymbol{0}$ and depends on the total mass M.

Reduced mass

- Define the **Reduced Mass**

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \qquad \frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$

- Since $\mu M = m_1 m_2$, multiply the relative equation by μ.

$$\boxed{\mu \boldsymbol{r}'' = -G m_1 m_2 \frac{\boldsymbol{r}}{r^3}}$$

$\boldsymbol{r}$ obeys this one-body central-force law with reduced mass $\mu > 0$.

Reconstructing both positions

- Substitute $\boldsymbol{r}_2 = \boldsymbol{r}_1 + \boldsymbol{r}$ into $M\boldsymbol{c} = m_1\boldsymbol{r}_1 + m_2\boldsymbol{r}_2$.

$$M\boldsymbol{c} = M\boldsymbol{r}_1 + m_2\boldsymbol{r}$$

- Solve for both positions.

$$\boxed{\boldsymbol{r}_1 = \boldsymbol{c} - \frac{m_2}{M}\boldsymbol{r}} \qquad \boxed{\boldsymbol{r}_2 = \boldsymbol{c} + \frac{m_1}{M}\boldsymbol{r}}$$

- In the center-of-mass frame, the bodies lie on opposite sides of the origin, with

$$\frac{\|\boldsymbol{r}_1\|}{\|\boldsymbol{r}_2\|} = \frac{m_2}{m_1}$$

Energy and reduced mass

- Differentiate the reconstructed positions.

$$\boldsymbol{r}_1' = \boldsymbol{c}' - \frac{m_2}{M}\boldsymbol{r}' \qquad \boldsymbol{r}_2' = \boldsymbol{c}' + \frac{m_1}{M}\boldsymbol{r}'$$

- Expand the kinetic energy. The mixed terms cancel.

$$\begin{aligned} T &= \tfrac{1}{2}m_1\|\boldsymbol{r}_1'\|^2 + \tfrac{1}{2}m_2\|\boldsymbol{r}_2'\|^2 \\ &= \tfrac{1}{2}M\|\boldsymbol{c}'\|^2 + \tfrac{1}{2}\mu\|\boldsymbol{r}'\|^2 \end{aligned}$$

- With $U = -Gm_1m_2/r$, the relative energy is

$$E_{\text{rel}} = \tfrac{1}{2}\mu\|\boldsymbol{r}'\|^2 - \frac{Gm_1m_2}{r}$$

Initial conditions and first-order form

- Transform initial positions and velocities by the same formulas.

$$\begin{aligned} \boldsymbol{c}(0) &= \frac{m_1 \boldsymbol{r}_1(0) + m_2 \boldsymbol{r}_2(0)}{M} \\ \boldsymbol{c}'(0) &= \frac{m_1 \boldsymbol{r}_1'(0) + m_2 \boldsymbol{r}_2'(0)}{M} \\ \boldsymbol{r}(0) &= \boldsymbol{r}_2(0) - \boldsymbol{r}_1(0) \\ \boldsymbol{r}'(0) &= \boldsymbol{r}_2'(0) - \boldsymbol{r}_1'(0) \end{aligned}$$

- For $\boldsymbol{v} = \boldsymbol{r}'$, the relative state $\boldsymbol{y} = (\boldsymbol{r}, \boldsymbol{v})$ satisfies

$$\boldsymbol{y}' = \left(\boldsymbol{v}, -GM \frac{\boldsymbol{r}}{\|\boldsymbol{r}\|^3} \right)$$

Earth–Moon masses

- Let body 1 be Earth and body 2 be the Moon. NASA lists the masses [2, 3].

$$m_1 = m_E = 5.9722 \times 10^{24} \text{ kg}$$
$$m_2 = m_M = 7.346 \times 10^{22} \text{ kg}$$

- The total mass is

$$M = m_E + m_M \approx 6.0457 \times 10^{24} \text{ kg}$$

- Treat Earth and the Moon as isolated point masses for this example; omit solar forces, tides, and nonspherical gravity.

Earth–Moon reduced mass

- Substitute the NASA mass values into the reduced-mass formula [2, 3].

$$\mu = \frac{(5.9722 \times 10^{24})(7.346 \times 10^{22})}{5.9722 \times 10^{24} + 7.346 \times 10^{22}} \text{ kg}$$

$$\boxed{\mu \approx 7.2567 \times 10^{22} \text{ kg}} \qquad \frac{\mu}{m_M} \approx 0.98785$$

- The reduced mass is about 1.22% smaller than the Moon's mass.

Earth–Moon center-of-mass frame

- The same masses give the position fractions [2, 3].

$$\frac{m_M}{M} \approx 0.012151 \qquad \frac{m_E}{M} \approx 0.98785$$

- Set $\boldsymbol{c} = \mathbf{0}$ and let $\boldsymbol{r}$ point from Earth to the Moon.

$$\boldsymbol{r}_E \approx -0.012151\boldsymbol{r} \qquad \boldsymbol{r}_M \approx 0.98785\boldsymbol{r}$$

- Both bodies move about their common center of mass.
- Their relative motion satisfies

$$\boldsymbol{r}'' = -G(m_E + m_M)\frac{\boldsymbol{r}}{\|\boldsymbol{r}\|^3}$$

References I

[1] NASA Earth Observatory.
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[2] David R. Williams.
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[3] David R. Williams.
Moon Fact Sheet.
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References II